

Week 2: Technology Research and Tool Finalization

Overview:

This week was spent exploring and validating the tools chosen in Week 1, ensuring they meet the technical requirements of our navigation system. The aim was also to start basic hands-on testing with these tools to prepare the team for active development.

Progress:

- Conducted deep-dive research on ARCore's capabilities and SDK integration methods.
- Explored Firebase features including Firestore, Realtime Database, and Firebase Authentication.
- Evaluated Android Studio plugin compatibility and sceneform support for AR.
- Assessed Figma for creating low and high-fidelity wireframes.
- Confirmed Java as the main development language for Android due to existing team expertise.
- Studied pathfinding algorithms and finalized the use of A* for shortest path computation.
- Created sample demos using Firebase for login and basic data storage.
- Developed internal documentation for how the selected tools will integrate into the overall architecture.

Summary:

Tool selection and research concluded successfully. The team is now confident in using the required technologies. A clear map of tool-function allocation has been drawn up for reference during development.

Next Week and Beyond:

Week 3 will begin the design phase with UI wireframing in Figma, targeting key workflows like station overview, facility search, and AR launch interface.